

LEMON-Factor: Bidirectional Factor-Level Semantic Alignment for Graph–Text Pairs

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Abstract. Graph–text evaluation should not stop at checking whether the right entity names appear in a fluent sentence. A faithful verbalization must preserve the relation roles, direction, and semantic commitments encoded by graph predicates; a reconstructed graph must recover the same information from text. We propose LEMON-FACTOR, a bidirectional diagnostic framework that decomposes predicates into weighted semantic factors and evaluates both $\text{KG} \rightarrow \text{Text}$ coverage and $\text{Text} \rightarrow \text{KG}$ recoverability. The method occupies a middle ground between whole-triple factuality checks and text-driven atomic-fact metrics: it keeps the source graph as the object of evaluation, but opens each predicate into auditable components. On WebNLG, Forward LEMON-FACTOR reaches 0.7915 and Reverse LEMON-FACTOR reaches 0.7789, while a MINE-STYLE comparison shows that node information is much easier to recover than edge information. Controlled perturbations confirm the diagnostic value of the factor view: deleting relation evidence reduces Forward LEMON-FACTOR by 0.211 while exact entity-label coverage changes by only 0.002. The same inventory format is then applied to DrugProt and BC5CDR, where biomedical predicates make role, direction, polarity, and causality explicit. A three-judge LLM reliability probe gives a complementary view: model judges often recover residual factors from context, whereas deterministic perturbations measure intended semantic damage. The study supports LEMON-FACTOR as an interpretable cross-domain framework for diagnosing relation-level graph–text alignment; it is not a substitute for expert human validation.

Keywords: Graph-to-text evaluation · Semantic alignment · Knowledge graphs · WebNLG · Biomedical relation extraction · Predicate decomposition

1 Introduction

Knowledge graphs express information through typed relations. In an edge (s, p, o) , the predicate p is not a decorative label between two entity names: it determines the roles of the arguments and the relation that should be preserved in text. The same problem appears in reverse. If a verbalization is converted back into a graph, a useful reconstructed edge must recover the predicate meaning, not only the two entities.

This distinction is easy to miss in graph-text evaluation. A sentence can mention *Alan Bean* and *Wheeler, Texas* without expressing that Wheeler is his place of birth. A biomedical abstract can contain the same chemical and protein names while changing inhibition into activation. Standard WebNLG evaluation uses reference metrics such as BLEU, METEOR, chrF++, TER, BERTScore, and BLEURT, as well as human criteria such as data coverage, relevance, correctness, and fluency [1–3]. These signals are useful, but they usually do not say which part of a predicate was preserved, weakened, or lost.

We address this gap with LEMON-FACTOR, a source-driven diagnostic framework for graph-text alignment. The method decomposes each predicate into typed, weighted semantic factors. For example, *birthPlace* includes a person-like subject, a biographical relation, and a place-like object. A DrugProt inhibition predicate includes a chemical actor, a gene/protein target, negative regulation, directionality, and evidence form. The score asks whether these factors are expressed in text or recovered in a reconstructed graph. It therefore gives partial credit when entities survive but relation meaning does not, and it produces a trace of the missing factors.

The framework is bidirectional. In the forward direction, $\text{KG} \rightarrow \text{Text}$, LEMON-FACTOR evaluates whether the text covers the semantic factors of the input graph. In the reverse direction, $\text{Text} \rightarrow \text{KG}$, it evaluates whether a reconstructed graph preserves the source predicate factors. We also implement a MINE-STYLE node/edge baseline inspired by KGGen’s Measure of Information in Nodes and Edges (MINE), which evaluates text-to-KG information retention through recovered nodes and relations [18]. This baseline is used as a transparent comparison point, not as a reproduction of the full KGGen benchmark.

The paper asks six research questions. **RQ1:** Does factor-level coverage capture graph-to-text semantic preservation beyond entity-label overlap? **RQ2:** Does the same factor representation support both forward coverage and reverse recoverability? **RQ3:** How does factor-level recoverability relate to node/edge information retention in a MINE-style baseline? **RQ4:** Do the metrics behave predictably under meaning-destroying and meaning-preserving perturbations? **RQ5:** Does the deterministic factor format produce useful perturbation diagnostics on biomedical graph-text data? **RQ6:** How do independent LLM factor judges compare with deterministic perturbation labels?

The contributions follow from this progression: a bidirectional factor-level formulation for graph-text semantic alignment; fixed factor inventories for WebNLG, DrugProt, and BC5CDR; reproducible forward, reverse, perturbation, and ablation analyses; a MINE-style node/edge comparison; and an exploratory three-judge LLM reliability probe. The study is diagnostic rather than leaderboard-oriented. Its purpose is to show when entity, triple, node/edge, and factor-level views disagree, and what those disagreements reveal about relation-level semantic preservation.

2 Related Work

2.1 RDF-to-text benchmarks and reference metrics

The WebNLG Challenge established RDF-to-text generation as a benchmark task in which systems map DBpedia-derived RDF triples to natural language [1]. The WebNLG+ shared task extended the setting to bilingual and bidirectional generation and included both automatic and human evaluation [2]. The GEM benchmark further argued for broader evaluation practice and challenge sets for NLG evaluation [3]. The standard WebNLG automatic stack includes reference-based metrics such as BLEU, METEOR, chrF++, TER, BERTScore, and BLEURT [4–6]. These metrics produce useful holistic similarity signals, but they do not localize errors to a source predicate or to a semantic role inside the predicate.

2.2 Source-aware coverage and factuality metrics

A second line of work evaluates generated text against input data. PARENT weights n-gram precision and recall by evidence in the source table and reference text [7]. Data-QuestEval adapts question-answering evaluation to data-to-text semantic coverage without requiring a reference [8]. AlignScore learns a unified alignment function for factual consistency across several supervision sources [9]. The closest graph-to-text predecessor is FactSpotter, which decides whether a given triple (s, p, o) is present in a generated text and reports strong correlations with human judgments for data correctness, coverage, and relevance [10]. LEMON-FACTOR complements this line by moving from a binary whole-triple decision to a weighted decomposition of the predicate into semantic factors.

2.3 Atomic facts, semantic graphs, and directionality

Atomic factuality metrics decompose generated text into verifiable claims. FActScore evaluates long-form text by decomposing it into atomic facts and checking support against a knowledge source [11]; VERISCORE extends claim verification for long-form factuality [12]. This direction is text-driven: the output text is decomposed into claims. LEMON-FACTOR is source-driven: the source predicate is decomposed into factors that can be checked in text or in a reconstructed graph.

Semantic graph metrics such as Smatch compare graph structures, especially AMR graphs, by maximizing overlap under node alignment [13]. S²match and Smatch++ add softer or more standardized graph comparison variants [14, 15]. These metrics are important for parser evaluation, but they presuppose graph representations on both sides. LEMON-FACTOR instead evaluates factor evidence from graph-text pairs and only uses reconstructed graphs in the reverse experiment.

2.4 Text-to-KG extraction and MINE

Recent text-to-KG work has renewed interest in evaluating whether extracted graphs retain useful information from plain text. KGGen introduces a language-model-based text-to-KG generator and releases MINE, Measure of Information in Nodes and Edges, as a benchmark for testing whether an extractor produces useful KGs from text [18]. Our MINE-STYLE baseline is inspired by this node/edge information split. It is not a reproduction of the KGGen benchmark, which targets KG extractors on plain text; rather, it adapts the idea to WebNLG graph-text pairs so that Text→KG recoverability can be compared with Reverse LEMON-FACTOR.

2.5 Behavioral testing and metric calibration

Evaluation metrics should be tested not only by correlation with human judgments, but also by whether they respond correctly to controlled changes. CheckList introduced behavioral tests for NLP models, including invariance tests, where label-preserving perturbations should not change predictions, and directional expectation tests, where perturbations should change predictions in a prescribed direction [19]. Perturbation CheckLists for NLG metrics apply the same idea to automatic metric evaluation [20]. He et al. use synthetic stress tests to expose blind spots of model-based NLG metrics [21]. Our calibration follows this methodology: relation deletion is a directional test, while entity aliasing and predicate paraphrase templates are invariance-style checks whose quality is explicitly marked.

Overall, existing metrics provide sentence-level similarity, source-aware coverage, binary triple checking, text-derived atomic facts, or node/edge retention. LEMON-FACTOR targets the missing layer between a whole predicate label and a full graph: weighted factors inside source predicates, used in both KG→Text and Text→KG directions.

3 Method

3.1 Problem setting

Let a graph-text pair be (G, T) , where G is a set of directed edges $e = (s, p, o)$ and T is a verbalization. The forward task asks whether the text covers the information in the graph. The reverse task asks whether a graph reconstructed from the text, denoted $\hat{G}(T)$, preserves the information in the original graph. LEMON-FACTOR uses the same predicate-level semantic representation in both directions (Fig. 1).

3.2 Predicate factorization

Each predicate p is mapped to a weighted decomposition:

$$D(p) = \{(f_i, r_i, w_i)\}_{i=1}^n,$$

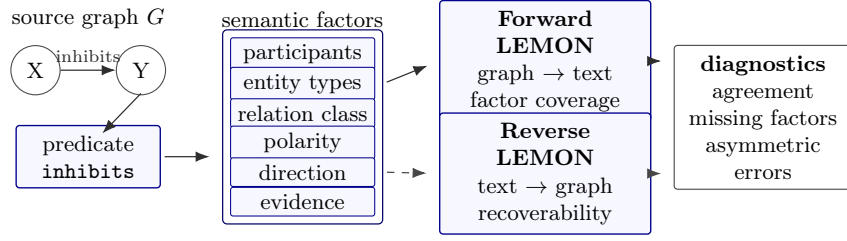


Fig. 1. Predicate-factor evaluation for graph-text alignment. A source predicate is decomposed into semantic factors, which are checked as forward graph-to-text coverage and reverse text-to-graph recoverability. Solid arrows denote forward evidence checks; dashed arrows denote reverse recovery checks.

where f_i is a semantic factor, r_i is its role, and $w_i > 0$ is a salience weight. Roles include *predicate meaning*, *subject domain*, *object domain*, *value domain*, and modifier or background roles. The schema uses broad factors such as *person*, *place*, *organization*, *time*, *quantity*, and relation factors such as *biographical relation*, *location relation*, *part-whole relation*, *political relation*, and *creative relation*. This representation is intentionally coarser than a full ontology but richer than a single predicate label.

For example, *birthPlace* receives a person-like subject factor, a biographical predicate factor, and a place-like object factor. In the biomedical inventories, *chemical_inhibits_gene_or_protein* receives chemical and gene/protein role factors, an inhibition factor, negative polarity, directionality, and evidence-form factors. A text that mentions both entities but omits the relation cue should therefore receive partial credit, not full coverage. This is the key distinction from whole-triple binary scoring.

3.3 Forward LEMON: KG→Text coverage

For an edge e , text T , and component (f_i, r_i, w_i) , an evidence function $E(f_i, r_i, e, T) \in [0, 1]$ estimates whether the component is expressed. In the deterministic implementation, entity and domain factors are supported by normalized label or token evidence, while relation factors are supported by predicate cues and factor cues. The edge-level forward score is:

$$S_F(e, T) = \frac{\sum_i w_i E(f_i, r_i, e, T)}{\sum_i w_i}.$$

Example-level and corpus-level scores are averages over edges with available decompositions. The implementation stores covered and missing factors for each edge, so the scalar score is accompanied by a diagnostic trace.

3.4 Reverse LEMON: Text→KG recoverability

The reverse direction first constructs a deterministic reconstructed graph $\hat{G}(T)$ from the text using entity-label evidence and predicate lexical cues. Reverse

LEMON-FACTOR then compares each gold edge with its best recovered counterpart. Subject and object recovery contribute through node/domain evidence; predicate recovery contributes through overlap between the gold predicate’s factors and the recovered edge evidence. The reverse score therefore asks whether the source predicate’s semantic factors survive the text-to-graph reconstruction.

The current reconstructed graph is a lexical pseudo-extractor, not a full KG extraction model. This choice makes the experiment reproducible and keeps the focus on metric behavior rather than extractor quality. Future work can replace $\hat{G}(T)$ with KGen, GraphRAG-style extraction, OpenIE, or domain-specific extractors.

3.5 MINE-style node/edge baseline

To compare factor-level recoverability with node/edge information retention, we implement a MINE-STYLE baseline inspired by KGen’s MINE. Each gold triple is treated as a fact, and the reconstructed graph is searched for matching or related nodes and edges. The deterministic score separates node information, edge information, and a composite partial-credit node/edge score. We report it as a local baseline rather than a reproduction of the original benchmark.

3.6 Controlled perturbation calibration

A metric that measures semantic alignment should degrade under meaning-destroying perturbations and remain comparatively stable under meaning-preserving perturbations. We therefore create deterministic perturbations of both text and reconstructed graphs. Text-side destructive perturbations include relation phrase deletion, object swaps, and predicate swaps. Graph-side destructive perturbations include node drops, edge drops, and predicate swaps. Preserving perturbations include alias-like rewrites and template paraphrases, which are flagged as clean or weak-template depending on their reliability. All perturbed files and operation-level manifests are saved, enabling audit of each injected change.

4 Data and Experimental Resources

4.1 WebNLG sample

The WebNLG experiments use a stratified English sample built from RDF-to-text examples. The train split contains 200 examples and the development split contains 100 examples; both cover 16 WebNLG categories. The development set contains 308 graph edges. The sample is small by design: it keeps the complete scoring pipeline auditable while still exposing relation-level errors that are hidden by entity overlap.

4.2 Biomedical transfer resources

The biomedical transfer checks use two relation extraction corpora converted into the same graph-text JSONL format. DrugProt is a BioCreative VII resource for chemical-gene/protein relations with 13 relation types and 24,526 manually annotated relations [16]. In the current processed subset, the train split contains 500 abstracts, 3588 edges, 12 observed predicate types, and an average text length of 1729.5 characters. BC5CDR is a BioCreative V corpus of 1500 PubMed articles with chemical and disease annotations and 3116 chemical-disease interactions [17]. The converted BC5CDR train and development splits each contain 500 documents, with 985 and 961 CID edges respectively. These resources are used as transfer diagnostics only; we do not train or claim a new biomedical relation extraction system.

4.3 Predicate-factor inventories

The experiments use fixed JSON predicate-factor inventories. Each inventory contains a controlled factor vocabulary and one weighted decomposition per predicate. WebNLG uses 121 predicate decompositions and covers all 119 predicate labels observed in the evaluation files. DrugProt contains 13 biomedical relation decompositions and covers the 12 relation types observed in the processed subset. BC5CDR is intentionally a single-relation setting: its inventory decomposes the document-level chemical-induced disease relation into chemical exposure, disease outcome, causality, document-level evidence, and chemical-to-disease direction. This makes BC5CDR useful for testing whether the same factor format transfers to a biomedical relation with causality and document-level evidence, not for testing predicate-label diversity.

The inventories are deterministic resources. Earlier LLM-generated decomposition candidates are kept in the repository as development material, but they are not treated as expert gold and are not used as the main evidence in the deterministic scoring tables below. The validator checks that coverage is complete for the current evaluation files and that the weights of each predicate decomposition sum to one.

5 Experiments

The experiments follow the six research questions as a single evaluation path. We first establish whether predicate-factor coverage adds information beyond lexical overlap, then test whether the same representation can be used in reverse, and finally stress the metrics with controlled perturbations across general-domain and biomedical data. Deterministic scoring runs use JSONL inputs and fixed JSON factor inventories. No LLM calls are made in the deterministic perturbation scoring reported in RQ5; LLM judging is isolated in RQ6.

5.1 RQ1: Forward factor-level semantic coverage

We first score WebNLG development texts using predicate decompositions and lexical cues. We then run an error-driven resource expansion: missing predicates are identified, rule-based or fallback decompositions are added, lexical cues are expanded, and coverage is recomputed. This measures how much of the initial low score comes from missing resources rather than from the text itself.

5.2 RQ2: Bidirectional factor recoverability

To test whether the factor representation is bidirectional, we reconstruct a graph from each WebNLG text with a deterministic lexical pseudo-extractor and compute Reverse LEMON-FACTOR. This score is compared with expanded Forward LEMON-FACTOR. A small forward-reverse gap suggests that the same factor schema can support both $\text{KG} \rightarrow \text{Text}$ coverage and $\text{Text} \rightarrow \text{KG}$ recoverability.

5.3 RQ3: MINE-style node/edge information retention

We compare Reverse LEMON-FACTOR with a MINE-STYLE baseline inspired by KGGen’s Measure of Information in Nodes and Edges. The deterministic variant reports node information, edge information, and a composite node/edge score. A previously locked 50-fact subset is also used to show how binary fact recoverability differs from a partial-credit node/edge composite. These values are reported as diagnostic baselines, not as a reproduction of the KGGen/MINE benchmark.

5.4 RQ4: Controlled perturbation calibration

We inject deterministic noise into text and reconstructed graphs. Destructive perturbations should reduce relevant scores; preserving perturbations should produce smaller changes. Text-side perturbations are used primarily to calibrate Forward LEMON-FACTOR, while graph-side perturbations are used primarily to calibrate Reverse LEMON-FACTOR and MINE-style recoverability.

5.5 RQ5: Cross-domain deterministic baseline scoring

The final perturbation files cover WebNLG, DrugProt, and BC5CDR. The operators keep the source graph fixed and change only the text, which lets us test whether a metric reacts to the intended semantic damage rather than to a change in the reference graph. The final scoring uses five operators: node deletion, relation-cue deletion, argument swap, polarity flip, and relation blur. Polarity flips are most informative for DrugProt and BC5CDR, where relation labels explicitly encode activation, inhibition, regulation, or causality.

We score the shared perturbation files with deterministic baselines. The goal is not to reproduce external systems such as FactSpotter or KGGen/MINE, but to provide local comparison points with the same inputs. We report entity recall,

Table 1. Forward LEMON-FACTOR before and after decomposition and cue expansion.

Metric	Initial	Expanded	Delta
Scored edges	140	308	+168
Missing decomposition rate	0.5455	0.0000	-0.5455
Predicate cue coverage	0.3149	0.6526	+0.3377
LEMON-FACTOR coverage	0.3683	0.7915	+0.4232
Low-coverage edges	191	41	-150

label match, a binary triple-style match, a MINE-STYLE node/edge harmonic score, a label-only LEMON variant, and the full deterministic LEMON-FACTOR proxy based on the fixed factor inventories and perturbation metadata. The same scoring layer is used for the compact ablation study.

5.6 RQ6: Exploratory LLM factor judging

To test whether deterministic perturbation labels agree with model-based semantic recovery, we run a small LLM reliability probe. We sample perturbed instances from WebNLG, DrugProt, and BC5CDR and ask three independent LLM judges to label each semantic factor as covered, partially covered, or absent. The judges are Gemini 2.0 Flash, GPT-4o-mini, and Llama-3.1-70B. The prompts use JSON-schema constrained outputs through OpenRouter. The probe is reported as diagnostic evidence, not as human validation.

6 Results

6.1 RQ1: Forward coverage and resource completeness

The first WebNLG run separates two sources of error: missing evaluation resources and missing information in the text. Before expansion, the factor dictionary covered only part of the development predicates, so 140 of 308 edges could be scored and the missing decomposition rate was 0.5455. After adding deterministic decompositions and lexical cues, all development edges had predicate decompositions and Forward LEMON-FACTOR increased from 0.3683 to 0.7915 (Table 1). The increase should not be read as a claim that all texts are faithful. It shows that factor-level evaluation requires explicit predicate resources before it can diagnose text quality.

The lexical controls make the distinction clearer. Exact entity-label overlap is already high and does not change after predicate-resource expansion; token cosine and Jaccard also remain fixed. The factor score changes because it is the only score in Table 2 that uses predicate decompositions and relation cues. In other words, the added signal is not a different way of counting entity mentions, but a check of relation-level content.

Table 2. Baseline comparison on the WebNLG development sample.

Method	Initial	Expanded	Delta
Exact entity-label overlap	0.9215	0.9215	0.0000
Predicate lexical cue coverage	0.3149	0.6526	+0.3377
Graph-text token cosine	0.7357	0.7357	0.0000
Graph-text token Jaccard	0.6233	0.6233	0.0000
LEMON-FACTOR coverage	0.3683	0.7915	+0.4232

Table 3. Bidirectional comparison on the WebNLG development sample.

Method	Direction	Factors	Retrieval	Score	Diagnostic level
Forward LEMON-FACTOR	KG→Text	yes	no	0.7915	factor/edge/example
Reverse LEMON-FACTOR	Text→KG	yes	no	0.7789	factor/edge/example
MINE-style composite node/edge	Text→KG	no	yes	0.8028	node/edge/fact
MINE-style LLM fact recoverability	Text→KG	no	yes	0.5200	fact/subgraph
Graph-text token cosine	KG↔Text	no	no	0.7357	corpus
Graph-text token Jaccard	KG↔Text	no	no	0.6233	corpus

6.2 RQ2: Bidirectionality

The same predicate factors can be used from graph to text and back from text to graph. The reverse experiment recovered 390 of 408 nodes and 196 of 308 edges. Reverse LEMON-FACTOR reached 0.7789, close to the expanded Forward LEMON-FACTOR score of 0.7915 (Table 3). The gap is small, but its interpretation is bounded by the setup: the reconstructed graph is produced by a deterministic lexical pseudo-extractor. The result therefore supports the representation, not the quality of a general-purpose extraction system.

6.3 RQ3: Relation information is harder than node information

The MINE-STYLE comparison explains why high graph-text overlap can still hide semantic loss. The deterministic composite node/edge score is 0.8028, but this aggregate is dominated by node recovery. On the locked 50-fact subset, node information is 0.9600, while edge information and deterministic fact recoverability are both 0.5400; the partial-credit composite is 0.7500. A separate LLM fact check gives a close score of 0.5200 and agrees with the deterministic fact decision on 0.9800 of the cases. The consistent pattern is that entity mentions are easier to preserve and recover than the relations that connect them.

6.4 RQ4: Perturbation calibration

The controlled perturbations turn this observation into a direct sanity check. If a relation phrase is deleted, the entity names should mostly remain, but a relation-aware metric should fall. This is what Table 4 shows: exact entity-label coverage changes by only -0.0020, while Forward LEMON-FACTOR drops by -0.2106. Predicate cue coverage and MINE fact recoverability fall even more sharply because they depend directly on the recoverability of relation evidence. The table therefore gives the central construct-validity signal: preserving the entities is not the same as preserving the predicate.

Table 4. Relation deletion sanity check. Delta is final minus baseline.

Metric	Baseline	Final	Delta
Exact label coverage	0.9215	0.9194	-0.0020
Forward LEMON-FACTOR	0.7915	0.5808	-0.2106
Predicate cue coverage	0.6526	0.2890	-0.3636
MINE fact recoverability	0.6364	0.2825	-0.3539

Table 5. Perturbation sensitivity as mean score drop. Larger values indicate that a metric reacts more strongly to the perturbation; unchanged or small drops mark blind spots rather than missing computations.

Perturbation	Entity	Triple	Node/edge	Label-only	LEMON-Factor
Node deletion	0.326	0.675	0.657	0.323	0.352
Edge deletion	0.215	0.655	0.637	0.482	0.281
Argument swap	0.214	0.540	0.521	0.321	0.413
Polarity flip	0.215	0.546	0.526	0.337	0.414
Relation blur	0.216	0.568	0.549	0.385	0.281

6.5 RQ5: Cross-domain deterministic scoring

The cross-domain perturbation files allow the same question to be asked on WebNLG, DrugProt, and BC5CDR: which metrics react when a controlled edit damages the meaning of the relation? Table 5 reports mean score drops, where larger values mean stronger sensitivity to the perturbation. Small or unchanged drops are not missing computations; they show that the metric is insensitive to that kind of damage.

The pattern is consistent with the intended diagnostic roles. Entity recall drops less under relation-focused edits because the entity strings often remain in the text. Triple-style and node/edge scores react more strongly because they require relation or edge evidence, but they still collapse the error into one coarse decision. LEMON-FACTOR separates the response by factor inventory. It reacts strongly to argument swap and polarity flip when those dimensions are encoded in the inventory, while the WebNLG polarity flip has little effect because most WebNLG predicates in this sample do not encode polarity. This is a feature of the formulation: the metric can only diagnose semantic dimensions that the inventory makes explicit.

The ablation table makes the role of decomposition visible. Removing participant-role factors produces the largest positive gain for the full model over the ablated variant, which means that role factors are the main source of sensitivity to argument and node-level damage. Removing evidence, polarity, or direction has smaller but interpretable effects, mostly in the biomedical setting. Label-only scoring can produce large drops when surface relation labels disappear, but it cannot say whether the damage concerns roles, direction, polarity, or evidence. The full factor score is therefore more diagnostic even when it is not always the harshest scalar score.

Table 6. Ablation sensitivity. Mean drop is the average perturbation response. Gain is full drop minus ablated drop; positive values mean that removing the component reduces sensitivity.

Variant	Mean drop Gain vs. full	
LEMON-full	0.320	0.000
Label-only	0.417	-0.097
Unweighted	0.310	0.010
Minus roles	0.185	0.135
Minus direction	0.310	0.010
Minus polarity	0.304	0.016
Minus evidence	0.290	0.030

Table 7. Exploratory LLM reliability probe. Judges are Gemini 2.0 Flash, GPT-4o-mini, and Llama-3.1-70B. Mean score treats covered, partial, and absent factor judgments as 1, 0.5, and 0; agreement is computed only on overlapping factor judgments.

Probe	Items	Judges	Mean score	Agreement
LLM factor judging	45	3	0.809	0.597

6.6 RQ6: LLM reliability probe

The LLM reliability probe tests a different question: whether model judges recover the same factors that deterministic perturbation labels mark as damaged. We ran three 45-item judge batches sampled from the prepared probe pool (135 judgments) with Gemini 2.0 Flash, GPT-4o-mini, and Llama-3.1-70B. Each judge saw the source edge, factor inventory, and perturbed text, and returned JSON-schema constrained labels: covered, partial, or absent.

Table 7 shows that LLM factor scores remain relatively high. The mean judged factor score is 0.809, while agreement with deterministic perturbation labels is moderate, and pairwise agreement on overlapping factor judgments is 0.597. This does not invalidate the deterministic perturbations. It shows that the two views measure different things. The perturbation layer records intended damage introduced by a controlled edit; the LLM judges estimate how much of the original factor remains recoverable from the residual context. This distinction is most useful for the later interpretation of limitations and future work: factor-level evaluation can expose the damage, while semantic judging can reveal when context still makes the damaged relation inferable.

7 Error Analysis

The experiments reveal two recurring error classes. The first is resource incompleteness: predicates without decompositions or with weak lexical cues cannot be scored reliably. The error-driven expansion step was designed to expose and

reduce this problem. After expansion, all 308 development edges had decompositions, but fallback decompositions remain less informative than expert-designed ones.

The second class is relation recoverability. Node information is high because WebNLG verbalizations often repeat entity labels, but edge information is substantially lower. This pattern appears in the reverse experiment, in the MINE-style node/edge gap, and in the relation-deletion perturbation. It suggests that the difficult part of graph-text alignment is not usually naming entities, but preserving and detecting the relation that links them.

The factor-level trace is useful here. A low LEMON-FACTOR can be traced to a predicate, a factor role, a missing cue, or a weak subject/object evidence signal. This is the main practical advantage of the framework: it turns a scalar score into a debugging signal for datasets, generators, and extraction pipelines.

8 Limitations

The study does not include expert human validation. The factor inventories are deterministic resources based on predicate labels, schema constraints, rules, and development-time candidates. The LLM probe adds a semantic recoverability check, but model judgments are not a replacement for expert annotation. Prior LLM-as-judge work shows that model judges can scale evaluation but can also exhibit preference and reasoning biases; future work should add expert factor annotations and inter-annotator agreement [22, 23].

The deterministic evidence layer is also limited. It can find labels and known cues, but it can miss implicit relations, long-range entailment, and paraphrases outside the cue dictionary. A stronger implementation should add embedding, entailment, or QA-based evidence while preserving the factor-level audit trail.

The reverse pipeline uses a lexical pseudo-extractor rather than a full text-to-KG system. This makes the experiment controlled, but it also means that the reverse scores should be interpreted as metric-behavior diagnostics. Similarly, the MINE-style baseline adapts KGGen’s node/edge information idea to graph-text pairs; it is not a reproduction of the original MINE benchmark.

Finally, the perturbations are deterministic and auditable, but still template-like. They support construct-validity checks for relation deletion, argument swap, and polarity flip, but they do not prove robustness to natural paraphrase. Drug-Prot and BC5CDR are used to test transfer of the factor format to biomedical roles, directionality, polarity, and causality; we do not claim state-of-the-art biomedical relation extraction performance.

9 Conclusion

LEMON-FACTOR evaluates graph-text alignment by opening relation predicates into semantic factors. This changes the unit of diagnosis. Instead of asking

only whether a triple is present or whether two entity names overlap, the framework asks which roles, relation meanings, directions, polarity cues, and evidence factors survive from graph to text and back to graph.

The experiments show why this level of analysis is useful. On WebNLG, the same factor representation supports both forward coverage and reverse recoverability. The MINE-style comparison shows that node information is much easier to recover than edge information, and the relation-deletion sanity check shows that entity-label coverage can remain almost unchanged while factor-aware relation coverage drops substantially. Across WebNLG, DrugProt, and BC5CDR, the deterministic perturbation analysis further shows that the score reacts to the semantic dimensions represented in the inventory: roles, direction, polarity, causality, and evidence.

The ablation and LLM reliability analyses refine this conclusion. Role factors contribute most to sensitivity under controlled damage, while label-only scoring remains less interpretable because it cannot say which part of the predicate was lost. The LLM judges provide a complementary view: they often recover factors from residual context even when a deterministic perturbation marks them as damaged. This does not weaken the factor approach; it clarifies its use. Deterministic factor scoring measures intended, auditable semantic damage, while model-based judging can reveal residual recoverability.

Taken together, the results position LEMON-FACTOR as a diagnostic layer between whole-triple factuality metrics and text-driven atomic-fact evaluation. Its value is not that it replaces existing metrics, but that it explains relation-level failures that coarser scores collapse into a single number. The next step is to replace provisional inventories with expert-validated factor annotations and to connect the same audit trail to stronger paraphrase and text-to-graph evidence models.

References

1. Gardent, C., Shimorina, A., Narayan, S., Perez-Beltrachini, L.: The WebNLG Challenge: Generating Text from RDF Data. In: INLG, pp. 124–133 (2017). <https://doi.org/10.18653/v1/W17-3518>
2. Castro Ferreira, T., et al.: The 2020 Bilingual, Bi-Directional WebNLG+ Shared Task: Overview and Evaluation Results. In: Proceedings of the 3rd International Workshop on Natural Language Generation from the Semantic Web (2020)
3. Gehrmann, S., Adewumi, T., Aggarwal, K., et al.: The GEM Benchmark: Natural Language Generation, its Evaluation and Metrics. In: Proceedings of the 1st Workshop on Natural Language Generation, Evaluation, and Metrics, pp. 96–120 (2021). <https://doi.org/10.18653/v1/2021.gem-1.10>
4. Papineni, K., Roukos, S., Ward, T., Zhu, W.-J.: BLEU: A Method for Automatic Evaluation of Machine Translation. In: ACL, pp. 311–318 (2002)
5. Zhang, T., Kishore, V., Wu, F., Weinberger, K.Q., Artzi, Y.: BERTScore: Evaluating Text Generation with BERT. In: ICLR (2020)
6. Sellam, T., Das, D., Parikh, A.P.: BLEURT: Learning Robust Metrics for Text Generation. In: ACL, pp. 7881–7892 (2020)

7. Dhingra, B., Faruqui, M., Parikh, A., Chang, M.-W., Das, D., Cohen, W.W.: Handling Divergent Reference Texts when Evaluating Table-to-Text Generation. In: ACL, pp. 4884–4895 (2019). <https://doi.org/10.18653/v1/P19-1483>
8. Rebuffel, C., Scialom, T., Soulier, L., Piwowarski, B., Lamprier, S., Staliano, J., Scoutheeten, G., Gallinari, P.: Data-QuestEval: A Referenceless Metric for Data-to-Text Semantic Evaluation. In: EMNLP, pp. 8029–8036 (2021). <https://doi.org/10.18653/v1/2021.emnlp-main.633>
9. Zha, Y., Yang, Y., Li, R., Hu, Z.: AlignScore: Evaluating Factual Consistency with a Unified Alignment Function. In: ACL, pp. 11328–11348 (2023). <https://doi.org/10.18653/v1/2023.acl-long.634>
10. Zhang, K., Balalau, O., Manolescu, I.: FactSpotter: Evaluating the Factual Faithfulness of Graph-to-Text Generation. In: Findings of EMNLP, pp. 10025–10042 (2023). <https://doi.org/10.18653/v1/2023.findings-emnlp.672>
11. Min, S., Krishna, K., Lyu, X., Lewis, M., Yih, W.-T., Koh, P.W., Iyyer, M., Zettlemoyer, L., Hajishirzi, H.: FActScore: Fine-Grained Atomic Evaluation of Factual Precision in Long Form Text Generation. In: EMNLP, pp. 12076–12100 (2023). <https://doi.org/10.18653/v1/2023.emnlp-main.741>
12. Sun, H., et al.: VERISCORE: Evaluating the Factuality of Verifiable Claims in Long-Form Text Generation. arXiv:2406.19276 (2024)
13. Cai, S., Knight, K.: Smatch: An Evaluation Metric for Semantic Feature Structures. In: ACL (2013)
14. Opitz, J., Daza, A., Frank, A.: AMR Similarity Metrics from Principles. Transactions of the Association for Computational Linguistics 9, 522–538 (2021). https://doi.org/10.1162/tacl_a_00329
15. Opitz, J.: Smatch++: Standardized and Extended Evaluation of Semantic Graphs. In: Findings of EACL, pp. 1595–1608 (2023). <https://doi.org/10.18653/v1/2023.findings-eacl.118>
16. Miranda-Escalada, A., Mehryary, F., Luoma, J., et al.: Overview of DrugProt BioCreative VII track: quality evaluation and large scale text mining of drug-gene/protein relations. Database 2023, baad080 (2023). <https://doi.org/10.1093/database/baad080>
17. Li, J., Sun, Y., Johnson, R.J., et al.: BioCreative V CDR task corpus: a resource for chemical disease relation extraction. Database 2016, baw068 (2016). <https://doi.org/10.1093/database/baw068>
18. Mo, B., Yu, K., Kazdan, J., Cabezas, J., Mpala, P., Yu, L., Cundy, C., Kanatsoulis, C., Koyejo, S.: KGGen: Extracting Knowledge Graphs from Plain Text with Language Models. arXiv:2502.09956 (2025)
19. Ribeiro, M.T., Wu, T., Guestrin, C., Singh, S.: Beyond Accuracy: Behavioral Testing of NLP Models with CheckList. In: ACL, pp. 4902–4912 (2020). <https://doi.org/10.18653/v1/2020.acl-main.442>
20. Sai, A.B., Dixit, T., Sheth, D.Y., Mohan, S., Khapra, M.M.: Perturbation Check-Lists for Evaluating NLG Evaluation Metrics. In: EMNLP, pp. 7282–7296 (2021). <https://doi.org/10.18653/v1/2021.emnlp-main.575>
21. He, T., Zhang, J., Wang, T., Kumar, S., Cho, K., Glass, J.: On the Blind Spots of Model-Based Evaluation Metrics for Text Generation. In: ACL, pp. 12067–12097 (2023). <https://doi.org/10.18653/v1/2023.acl-long.674>
22. Liu, Y., Iter, D., Xu, Y., Wang, S., Xu, R., Zhu, C.: G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment. In: EMNLP, pp. 2511–2522 (2023). <https://doi.org/10.18653/v1/2023.emnlp-main.153>
23. Zheng, L., Chiang, W.-L., Sheng, Y., et al.: Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In: NeurIPS (2023)